

A Physics-Cost Neural-OT Upgrade of the Sliced-Wasserstein Autoencoder for Detector-Level $J/\psi \rightarrow \mu^+\mu^-$ Simulation

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Abstract

The OTUS sliced-Wasserstein autoencoder learns a stochastic detector response between truth-level dimuon four-vectors and CMS detector-level four-vectors without per-event paired simulation. The literal 2021 objective matches broad coordinate distributions but misses the ~ 30 MeV J/ψ mass shell. This paper reports two incremental upgrades: (i) a Run-D baseline adding pair-level sliced Wasserstein and a relative-mass cycle constraint, which reduces the simulation mass Wasserstein-1 from 0.95 GeV to 0.026 GeV; and (ii) a Run-E “Tier A” upgrade that replaces blind random slicing with a physics-cost Sinkhorn divergence and adversarially learned slicing directions, parameterizes the transport in cylindrical coordinates as a short stochastic residual flow, and selects checkpoints with hard mass-shape gates. Modern classifier two-sample, coverage, and per-event stochasticity diagnostics are introduced in place of marginal histogram checks alone. **[TODO: Abstract final numbers after Run E.]**

1 Introduction

Fast surrogate simulation of detector effects is a central application of generative models in high-energy physics [1, 2]. OTUS reformulates the problem as an unpaired transport between a physically interpretable latent space of generator-level particles, $\mathcal{Z}$, and detector-level observations, $\mathcal{X}$ [1]. Its core loss is

$$\mathcal{L}_{\text{OTUS}} = \mathbb{E}_{x \sim p(x)} \mathbb{E}_{z \sim p_E(z|x)} \mathbb{E}_{\tilde{x} \sim p_D(x|z)} \|x - \tilde{x}\|^2 + \lambda d_{\text{SW}}(p_E(x), p_D(z))$$

(1) optionally augmented by directional anchor terms. The sliced-Wasserstein term projects high-

dimensional particle vectors onto random one-dimensional axes; our measured failure mode is that a narrow resonance shell is nearly invisible to most of those axes.

This paper makes two contributions:

1. a systematic comparison of the literal OTUS objective, the Run-D pair-level extension, and the Virat-style stochastic residual baseline;
2. the Run-E “Tier A” method: physics-aware neural-OT costs, learned max-sliced-Wasserstein directions, a cylindrical residual-flow parameterization, mass-gated model selection, and modern generative diagnostics.

2 Related Work

OTUS builds on the sliced-Wasserstein autoencoder [3, 4]. Neural optimal transport, notably entropic regularization [5], Sinkhorn divergences [6], and Monge-gap regularizers, replaces random projection distances in modern unpaired-transport models. Flow matching and stochastic interpolants provide continuous-time conditional generative models [7, 8]; our cylindrical residual flow is a discrete, physics-constrained step in that direction. Fast detector simulation has moved from GANs [2] toward normalizing flows, diffusion models, and equivariant particle-cloud models [9]. For (unfolding) OmniFold is the current reference [10]; (1) our encoder remains a single-pass stochastic unfolding map and is not claimed to replace OmniFold.

3 Method

3.1 Baseline and Run D

The detector-level data are CMS 2012 DoubleMu-Parked opposite-sign dimuons in the J/ψ window. The Run-D objective extends 1 with a direct generator term and a pair-level SWD,

$$\mathcal{L}_{\text{RunD}} = \alpha(t) [d_{\text{SW}}(z, E(x)) + d_{\text{SW}}(x, D(z))] + \lambda_{\text{cyc}} \mathbb{E}[\|D(\log(m)) - \log(m)\|^2] + \lambda_{\text{H006C}} \left(\frac{m_{\tilde{x}} - m_x}{m_x} \right) \quad (2)$$

3.2 Run E Tier A components

Physics-cost Sinkhorn and Monge gap. We define a 14-dimensional standardized ground feature vector,

$$u(p) = [\log p_{\text{T}}^-, \eta^-, \sin \phi^-, \cos \phi^-, \log p_{\text{T}}^+, \eta^+, \sin \phi^+, \cos \phi^+, \log m, \log p_{\text{T}}^{\mu\mu}, y, \cos \Delta\phi, \sin \Delta\phi, \Delta\eta] \quad (3)$$

and use a weighted squared cost $c(a, b) = \sum_i w_i (u_i(a) - u_i(b))^2$. The distribution distance is the debiased Sinkhorn divergence; a Monge-gap proxy with barycentric projection is available as a regularizer.

Max-SW. Instead of random slicing directions, a small direction set is updated adversarially to maximize the sliced distance on each minibatch, with an orthogonality penalty, and the generative model minimizes the same distance.

Cylindrical residual flow. Encoder and decoder are short flows in cylindrical coordinates. At each of K steps a physics-conditioned network predicts a bounded mean residual, a Gaussian core scale, and a Student- t tail scale; azimuthal coordinates are transported on S^1 and energies are reconstructed on the muon mass shell.

Mass-gated selection. A checkpoint is eligible only if the generated mass satisfies hard gates on W_1 , KS, window fraction, and mean shift. The selected-score is set to infinity for ineligible checkpoints.

Modern diagnostics. We report classifier two-sample test AUC, fixed- z repeated-decoding stochasticity, and rank-based coverage for cycle and nearest-neighbor pseudo-pairs.

Table 1: Run D measured test-split mass metrics.

Checkpoint	W_1 [GeV]	KS
three-term baseline	0.946	0.764
Run D best (epoch 270)	0.026	0.221
Run D final (epoch 300)	0.061	0.296

4 Experimental Setup

Training uses $\sim 580,000$ CMS dimuon events and 75,904 MG5 truth events from the rebuilt ptj=5 prior (80/10/10 splits, 20% CMS cap). Models are trained on Apple-silicon MPS. Evaluation is performed on the untouched test split. **[TODO: Add exact software versions and commit hashes.]**

5 Results

5.1 Distribution-level results

[TODO: Insert Run E mass ratio/residual figures from plots best.] Expected reporting table: W_1 , KS, peak position, width ratio, window fraction, C2ST AUC, and central coverage for $D(z)$, $E(x)$, and $D(E(x))$.

5.2 Stochasticity and coverage

[TODO: Insert fixed- z per-event width table and rank-uniformity plots.]

5.3 Ablations

Planned single-variable ablations: Sinkhorn on/off, max-SW on/off, flow steps $K = 1, 2, 4$, mass gates on/off.

6 Discussion

Limitations: no per-event paired CMS truth is available, so conditional coverage is evaluated only through cycle and nearest-neighbor proxies; the skim exposes no isolation, impact-parameter, or pileup variables; and the method is demonstrated for a fixed two-particle final state. The generalization to variable particle multiplicity, full event objects, and Lorentz-equivariant representations remains future work.

7 Conclusion

[TODO: Final conclusion after Run E.]

References

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